

All Results — Master Reference

Every empirical result in one place. Framework: Xie et al. (2011) committed-minority tipping point ($p_c \approx 10\%$). Question: does it transfer to LLM societies, and how do homogeneity and scale modulate it? Total API spend $\approx$ \$16.

R1. Mechanical baseline validates the simulator

Pure automaton binary-agreement model reproduces the analytic tipping point. | N | measured p_c | |—|—| | 200 | 0.0929 | | 500 | 0.0936 | | 1000 | 0.0972 | | 2000 | 0.0983 | → converges to analytic $p_c = 0.0979$. Order parameter collapses sharply at p_c .

R2. LLM naming game — the tipping point transfers ($\sim 10\%$), bias-controlled

Capable models, neutral randomized labels + counterbalancing (intrinsic bias $h \approx 0$): | Lineage | p_c | |—|—| | DeepSeek | 0.084 | | Anthropic (haiku) | 0.093 | | Qwen-72B | 0.095 | | Meta (Llama-70B) | 0.101 | | OpenAI (gpt-4o-mini) | 0.118 | → All five lineages cluster at $\sim 10\%$, matching the analytic 0.0979. Homogeneous and heterogeneous compositions are indistinguishable (the game is a contagion model).

R3. Capability gate — naming-game rule-following by scale

Accuracy on the 3-case agreement rule (neutral labels, sharp prompt): | Model | Scale | Rule acc | |—|—|—| | llama3.2:3b / qwen2.5:3b / gemma2:2b | 2-4B | 33-77% ✖ | | deepseek-v3 | MoE | 88% | | llama-3.3-70b / qwen2.5-72b | 70B | 100% ✔ | | qwen3-235b | 235B | 98% | | gpt-4o-mini / claude-haiku / claude-sonnet / gpt-4o / gpt-5 | frontier | 100% (gpt-5-nano 33% — reasoning hurts) | → Below $\sim 30B$ the coordination primitive fails; the “scale effect” is a threshold.

R4. Error correlation — the illusion of cognitive diversity

Capable models (7 lineages, 40 hard Qs): mean $\rho = 0.53$, $N_{\text{eff}} = 1.7$ of 7. At-least-one-right 90% vs best single 81% → headroom +9%; 10% all-wrong. **Frontier (7 models incl. GPT-5, Claude-Opus, 100 hard Qs):** mean $\rho = 0.48$, $N_{\text{eff}} = 1.8$ of 7. Best single = claude-opus 92%; at-least-one-right 95% → headroom +2% (a dominant model erases the diversity benefit). → Even across independent lineages, LLMs fail on the *same* questions.

R5. Factual debate — heterogeneity does NOT net-win vs a strong monoculture

Setting	Accuracy
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claude-haiku solo	63%
homo claude-haiku ×6 (debate)	75% (self-consistency, +12)
hetero [haiku+gpt4o-mini+qwen72b]	72% (dilution)
claude-sonnet solo	81%
homo claude-sonnet ×3	86%
hetero matched [sonnet+qwen235b+llama70b] + judge	80% maj / 82% judge

→ Strong monoculture beats diverse pool; dilution > the small ρ -limited decorrelation.

R6. Case studies (same questions, $p=0$): decorrelation wins vs dilution losses

On 40 hard Qs, homo:claude-haiku vs hetero 3-lineage: **5 decorrelation-wins** (homo wrong → hetero right, e.g. TruthfulQA “drive across country”: 6 identical claude-haiku “A” answers vs gpt4o-mini+qwen72b correcting to “C”) vs **9 dilution-losses** (homo right → hetero dragged wrong). Net $-4 \rightarrow$ homo 75% > hetero 72%. (See CASE_STUDIES.md.)

R7. Competence band — weak models and the aggregator

Three sub-floor weak models (gemma2 33%, llama3b 34%, qwen2.5 38%; $\rho = 0.29$): | measure | one weak | one weak ×3 (homo) | three different weak (hetero) | |—|—|—|—| | majority-vote accuracy | 33% | 36% | **36%** (no gain) | | coverage (≥ 1 right) | 38% | $\sim 38\%$ | **60%** (+22%) | → Below the 0.5 floor, two weak = one weak by voting, but two diverse weak $\gg$ one by coverage. “Two weak beat one” holds for verify/select use cases, fails for voting.

R8. Value deliberation — heterogeneity WINS (no ground truth)

Moral dilemmas/fables, homo:claude-sonnet vs hetero:5-architectures: | Metric | homogeneous | heterogeneous | |—|—|—| | distinct considerations | 5.9 | **6.9** | | final stance spread (0 = false consensus) | **0.00** | **0.47** | | sway toward committed extremist (lower = resists) | 0.20 | **0.13** | → Homo collapses to *identical* stances with verbatim reasoning (groupthink); hetero preserves disagreement, raises more considerations, resists the extremist. (See DELIBERATION_DYNAMICS.md.)

R9. GovSim cooperation — capability gate + tipping generalizes to $\sim 25\%$

Model	$p=0$ (no defectors)	tipping
gpt-4o-mini	collapses (~ 4 mo)	— (below cooperation floor)
claude-sonnet	sustains 10/10 mo, yield 500 (sustainable max)	sustains $\leq 20\%$ defectors, collapses at 30%

→ Committed-minority tipping **generalizes to cooperation at ~25%** (Centola-like), distinct from the ~10% opinion threshold. Weak models can't cooperate at all.

R10. Cost vs quality — when cheap heterogeneous beats one expensive model

General (mixed, 100 Qs): cheap panel majority 78% (= best cheap single) vs opus 92% — blind voting can't reach opus. Coverage (verify) 91% $\approx$ opus. **Verifiable STEM (MMLU-Pro math/sci, 40 Qs):** | Approach | Accuracy | cost/1000Q | |—|—|—| | **claude-opus alone** | **81%** | ~\$11 | | gpt-4o / deepseek-v3 / qwen235b (each alone) | 48% / 56% / 68% | | | **cheap panel [3 models] + judge** | **81%** ✓ | ~\$5 | → On verifiable tasks a cheap diverse panel + checker **matches Opus at ~2× lower cost**. On non-verifiable tasks, use the best single affordable model. Decision tool (decision_tool.py): $C = 1 - (1 - \bar{a})^{N_{\text{eff}}}$; $R = a_{\text{best}} + v \cdot (C - a_{\text{best}})$; recommend cheap panel iff $R \geq a_{\text{expensive}}$ and cheaper. (See COST_QUALITY.md.)

Synthesis (answers to the professor's three questions)

- **RQ1 (homo/hetero):** monocultures more vulnerable to groupthink/common-mode (R8, R6), but heterogeneity's protection is bounded by ρ (R4), can backfire via dilution (R5), and is realized only by the right aggregator (R7, R10).
- **RQ2 (scale):** a capability gate / competence floor, not a smooth curve (R3, R9).
- **RQ3 (tipping points):** ~10% committed-minority threshold in opinion (R2) and ~25% in cooperation (R9), plus the Condorcet floor (R7).

Thesis: heterogeneity helps *conditionally* — gated by a competence floor, bounded by error correlation (LLMs are more correlated than they look), and realized only by aggregators that exploit decorrelation — but it wins cleanly on open-ended deliberation.

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